

12) $KE = \frac{1}{2} mv^2 = 5.0 \times 10^{-13} \text{ J}$
mass of proton = $1.67 \times 10^{-27} \text{ kg}$
 $v^2 = 5.988 \times 10^{14}$

acceleration = $v^2/r = 9.98 \times 10^{14} \text{ ms}^{-2}$
Answer D

13) a proportional to $1/r^2$